Hyungu Kim

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Education

Yonsei University Mar 2020 – Feb 2027 (expected)
B.S. in Electrical and Electronic Engineering *Seoul, Korea*
GPA: 3.93/4.30 (cumulative), 4.14/4.30 (major)

Publications

(*: equal contribution)

[P1] VGI-SKT: Competition Solution for the 2026 BEHAVIOR Challenge
Woosung Kim*, **Hyungu Kim***, Hoyong Kim*, Chunfei Ma, Seungjun Lee, Joonhyang Choi, Haebeom Jung, Hyunjoon Lee, Sunghyun Choi, Daeun Lee, Eunsung Cha, Seokmin Song, Junghwan Park, Jaesik Park
Technical Report, BEHAVIOR Challenge 2026 [report] [code] [Hugging Face]

[P2] Monocular 6-DoF Object Pose Tracking with Learned Depth and Matching Priors
Semin Kim, **Hyungu Kim**, Jongyoo Kim
APSIPA ASC 2026 [report]

Research Experience

SNU Visual & Geometric Intelligence Lab Oct 2025 – Present
Undergraduate Research Intern, Advisor: Prof. Jaesik Park

- Led the BEHAVIOR 2026 entry: G0.5 VLA + belief-graph CoT; 0.445 Q-score, 22.3% success **[P1]**.
- Designed the System-2 harness (LLM router + state machine over a VLA); won AgentBeats **[A1]**.

Yonsei Multimedia AI Lab Dec 2025 – Jun 2026
Research Intern, Advisor: Prof. Jongyoo Kim

- Built CAD-free 6-DoF tracking: generated-mesh pipeline, temporal-attention tracker, retraining **[P2]**.
- Showed MASt3R fails part-level matching on wide baselines; fixed it with SigLIP features.

SNU PI Lab Mar 2025 – Jun 2025
Research Intern, Advisor: Prof. Youngjae Yu

- Sim-to-real VLA navigation; waypoints generated from language + sketch instructions.

Honors and Awards

[A1] 1st Place (tie), AgentX–AgentBeats, Phase 2 — VGI-Player, Minecraft agent 2026
[A2] 1st Place, AgentX–AgentBeats, Phase 1 — agentified Minecraft (MCU) benchmark 2026
[A3] Academic Excellence Award, Yonsei University Spring 2021, Spring 2025

- Next-Generation Engineer Badge, IPESK
- Physics Expert Certification

Research Interests

Long term: *embodiment-agnostic intelligent collaboration* among humans, agents and robots. Key components:

- **Memory management** for long-horizon agents (e.g., **[P1]**)
- **Multi-agent collaboration** (e.g., **[A1]**, **[A2]**)
- **Whole-body control** (e.g., **[P1]**)
- **Physics & geometry understanding** (e.g., **[P2]**)

Currently:

- **Integrating and harnessing VLAs, WAMs and LLMs** (e.g., **[P1]**, **[A1]**)

Skills

Languages	Python, C++	**ML**	PyTorch, JAX, Hugging Face
Simulation	Isaac Sim, OmniGibson	**Serving & Infra**	vLLM, Docker, CUDA, Linux